

Protocol: Probe-Seq v.1.3

NOTE: If Probe-Seq is not successful for any reason, please check the Troubleshooting Guide at the end of the document.

NOTE: If you are interested in using nuclei from frozen tissue, please refer to the Appendix.

DAY 1**Tissue Dissociation (30 minutes)**

**Dissociation may differ for different tissues. This dissociation protocol is for the retina.

**If starting with already dissociated cells, start with step 12.

1. Prepare all solutions.

NOTE: Be sure that RNase inhibitor has been added to all solutions. Papain mix should be moved to 37°C to warm for ~15 minutes. HBSS/FBS and DMEM/BSA should be at room temperature. 4% PFA and Permeabilization Buffer should be placed on ice. 40% wHyb (warm 500 µL per sample) and Probe Mix should be warmed to 43°C.

2. Dissect retinas in HBSS or PBS at RT.
3. Place retina in a microcentrifuge tube and remove most of the PBS/HBSS without disturbing the retina tissue.
4. Add 400 µL of the pre-warmed (37°C) Papain Mix.
5. Incubate for 7 minutes at 37°C with no agitation.
6. Spin 600xg for 2.5 minutes at RT.
7. Remove supernatant and add 1 mL of HBSS/FBS without agitation.
8. Spin 600xg for 2.5 minutes at RT.
9. Remove supernatant and add 600 µL of DMEM/BSA.
10. Dissociate by triturating the tissue with a P1000 pipette. ~15 times in/out pipetting should be sufficient to dissociate into single cells. The solution should turn homogenous with no visible clumps.

NOTE: Avoid bubbles. If large chunks of debris or clumped cells are present at this stage, it might be advised to carefully remove them using a pipette or forceps.

11. Transfer homogenate to a 5-mL polypropylene tube (Thermo Fisher Scientific cat. #14-959-11A). Make sure to seal the cap by pushing in.

FISH (~1 hour hands on time (Day 1), O/N incubation, ~2 hours hands on time (Day 2))

12. Spin 600xg for 5 minutes at 4°C.

NOTE: If a temperature-controlled centrifuge is not available, RT spin is also acceptable. However, there may be slightly more RNA degradation.

13. Remove supernatant and add 1 mL of 4% PFA. Resuspend. Incubate for 15 minutes at 4°C with rocking.

NOTE: To minimize cell loss, ~50 -100 uL of liquid can be left at the bottom after each spin step.

14. Spin 2000xg for 5 minutes at 4°C.

NOTE: The centrifugation speed has changed to 2000 xg after fixation.

15. Remove supernatant and add 1 mL of Permeabilization Buffer. Resuspend. Incubate for 10 minutes at 4°C with rocking.

16. Spin 2000xg for 5 minutes at 4°C.

17. Remove supernatant and add 500 µL of pre-warmed (43°C) 40% wHyb. Resuspend. Incubate for at least 30 minutes at 43°C in the oven.

NOTE: If an *in situ* hybridization oven is not available (i.e. Bellco cat. #7930-01230), a water bath should suffice although it has not been tested in our hands.

18. Spin 2000xg for 5 minutes at RT.

NOTE: When changing liquids, it is critical to perform every step quickly to minimize change in temperature.

NOTE: It is crucial that the Probe Mix is at 43 °C when it is added to the sample. If the Probe Mix is added at a lower temperature, significant off-target binding of probes can occur causing unwanted background. This is also important for subsequent addition of wHyb solution for washes.

NOTE: From here, resuspension is not necessary until FACS. Resuspension will cause cells to get stuck onto the pipette tip.

19. Remove supernatant and add 100 µL of pre-warmed (43°C) Probe Mix. Incubate for at least 16 hours at 43°C in the oven.

-----O/N Incubation-----

DAY 2

NOTE: If RNA sequencing is not going to be performed for your samples and the RNA quality is not important after FACS, you can perform a longer incubation. Increasing the incubation time to 2 overnight incubations (~48 hours) has increased the staining index. However, the RNA quality after such long incubation has not been tested.

20. Prepare all solutions

NOTE: Be sure that RNase inhibitor has been added to all solutions

NOTE: Approximately 30 minutes before starting, you should warm the 40% wHyb (1mL/sample) and 2X SSC (1mL/sample) to 43 °C. It is critical that they reach 43 °C.

NOTE: At this point, another oven should be set to 37 °C for the fluorescent oligo hybridization and washes. You should warm the Fluorescent Oligo Mix and PBS (1.5mL/sample) to 37°C.

NOTE: If you are working with a single oven, you can pause after step 24. Remove the supernatant from the 2X SSC wash (leaving ~100uL and tightly cap to prevent desiccation) and switch the

oven temperature to 37 °C. Put the sample in the oven at 37°C and wait ~10 minutes for equilibration. Then, resume at step 25 by adding pre-warmed 37°C PBS.

21. Add 500 µL of pre-warmed (43°C) 40% wHyb and spin 2000xg for 5 minutes at RT.

NOTE: The pellet might become transparent after incubation in the 40% wHyb. Continue to leave ~100uL in the tube when removing supernatant to avoid disturbing the pellet.

22. Remove supernatant and add 500 µL of pre-warmed (43°C) 40% wHyb. Incubate for 15 minutes at 43°C.

23. Spin 2000xg for 5 minutes at RT.

24. Remove supernatant and add 1 mL of pre-warmed (43°C) 2x SSC. Incubate for 5 minutes at 43°C.

25. Spin 2000xg for 5 minutes at RT.

26. Remove supernatant and add 500 µL of pre-warmed (37°C) PBS.

27. Spin 2000xg for 5 minutes at RT.

28. Remove supernatant and add 100 µL of pre-warmed (37°C) Fluorescent Oligo Mix. Incubate for 10 minutes at 37°C.

29. Add 500 µL of pre-warmed (37°C) PBS and spin 2000xg for 5 minutes at RT.

30. Remove supernatant and add 500 µL of pre-warmed (37°C) PBS. Incubate for 5 minutes at 37°C.

31. Spin 2000xg for 5 minutes at RT.

NOTE: Before resuspending, pipette up and down with FACS Collection Solution (BSA/PBS) to minimize cells sticking inside the pipette tip.

32. Resuspend in 500 – 1000 µL of PBS depending on the number of cells. Keep tubes on ice.

33. Filter and proceed to FACS.

NOTE: Make sure to bring collection tubes (microcentrifuge tubes) with 500 µL of FACS Collection Solution per population.

FACS (~30 minutes for 1 million cells)

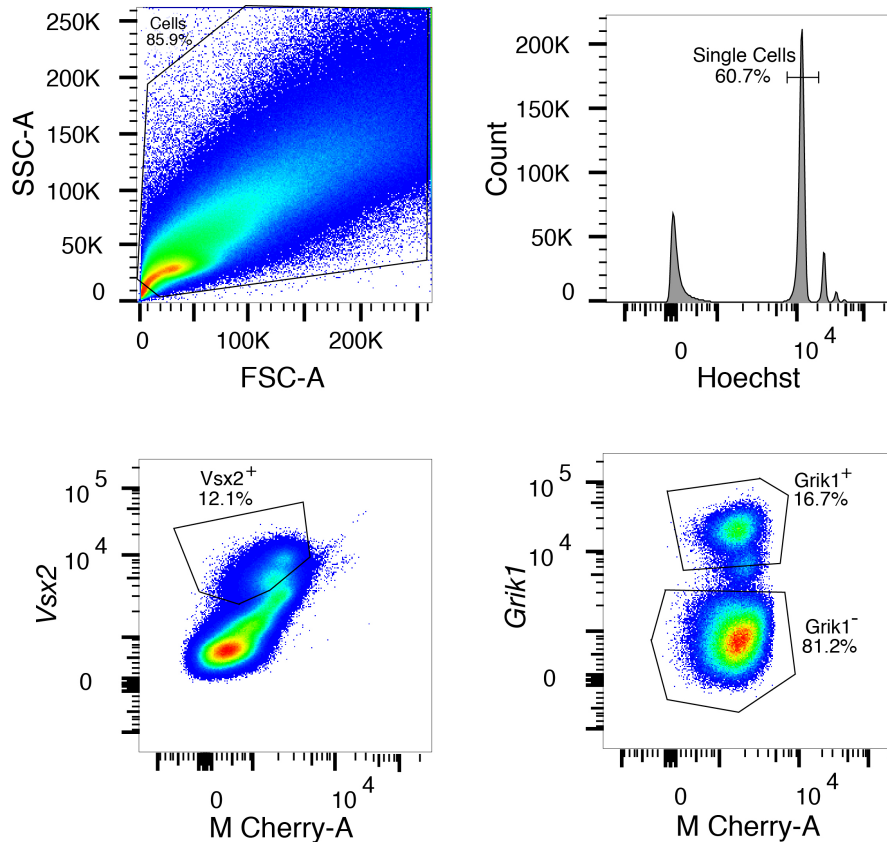
34. Gate based on Hoechst histogram, as shown below. This step will ensure that you get 2N nuclei. Make sure to NOT include 4N nuclei. You can also gate based on FSC/SSC for cells vs. debris (top left graph), but this step is likely unnecessary as debris will be negative for Hoechst.

NOTE: If your tissue contains polyploid cells (4N+ cells), it will be difficult to distinguish them from doublets or other cell clumps. In this case, it may be best to use the nuclei Probe-Seq protocol.

35. Gate using a 2D plot with appropriate wavelengths. The negative population will likely run diagonal. The positive population will be left- or right-shifted compared to the diagonal events (i.e. the $V_{sx}2^+$ cells below). Often, the separation will not be as obvious as GFP or well characterized cell surface markers. Thus, it's a good idea to run a negative control (fluorescent oligo only control).

NOTE: It is advisable to keep one laser channel unused as this can be used to determine autofluorescence (i.e. the M-Cherry channel in the graphs below). If there are events that are highly fluorescent in your channel of interest (i.e. the Grik1 channel below) AND the autofluorescence channel (i.e. M-Cherry), these events are likely autofluorescent in all channels. Thus, these events should not be collected.

36. FACS isolated cells are sorted into FACS Collection Solution in a microcentrifuge tube (start with 500 µL per population already in the tube) and kept at 4°C. This is NOT a stopping point.



Decrosslinking and RNA isolation (~4 hours)

37. Transfer the collected cells to a 5-mL polypropylene tube (with white caps).
38. Spin at 3000xg for 7 minutes at 4°C (or at RT if temp-controlled centrifuge is not available)

NOTE: Centrifugation at RT is also acceptable. Note the change in spin speed.

39. Remove as much supernatant as possible. Usually, there is ~40-50 µL left.
40. From the Recoverall RNA/DNA Isolation Kit (Thermo Fisher Scientific AM1975), mix 100 µL of Digestion Buffer and 4 µL of protease for each sample.
41. Add the Digestion Buffer/Protease Mix (104 µL). Resuspend.
42. Transfer the solution to a DNA LoBind microcentrifuge tube (Eppendorf cat. #Z666548) if available. Wrap the lid with parafilm.
43. Incubate at 50°C for 3 hours.

NOTE: This incubation time differs from the manufacturer's protocol.

44. After 3 hours, the samples can be stored at -80°C indefinitely or proceed to next steps according to the kit protocol.
45. Prep Isolation Additive/Ethanol mix (for 100 µL digest, 120 µL isolation + 275 µL 100% Ethanol. Usually, there is ~140 µL so adjust accordingly).
46. Add Isolation/Ethanol Mix and pipette to mix.
47. Add onto filter and spin 10000xg for 30 seconds at RT.
48. Discard flow-through. Wash with 700 µL of Wash 1. Spin 10000xg for 30 seconds at RT.
49. Discard flow-through. Wash with 500 µL of Wash 2. Spin 10000xg for 30 seconds at RT.

50. Discard flow-through. Spin 10000xg for 30 seconds at RT.
51. Prep DNase Mix (6 μ L 10x DNase Buffer, 4 μ L DNase, 50 μ L UltraPure Water).
52. Add 60 μ L of DNase Mix to center of cartridge. Close cap and incubate for 30 minutes at RT.
53. Add 700 μ L of Wash 1, incubate 1 minute at RT. Spin 10000xg for 30 seconds at RT.
54. Discard flow-through. Wash twice with 500 μ L of Wash 2.
55. Spin 10000xg for 1 minute to remove residual fluid.
56. Elute in ~17 μ L of UltraPure water (Use more water if lower concentration desired. Eluting with 17 μ L will leave you with ~12 μ L of RNA, which is enough for SMART-Seq v.4, Qubit, and/or BioAnalyzer).
57. Store RNA at -80°C.

NOTE: Qubit or BioAnalyzer can be used to estimate RNA concentration. If less than 10,000 cells, concentration may not be available by these methods. In this case, proceed directly to SMART-Seq v.4 using 9.5 μ L of RNA as the starting volume.

58. Proceed to the SMART-Seq v.4 protocol for cDNA synthesis and amplification. Depending on the number of cells, you will use 9.5 μ L of RNA for cDNA synthesis.

NOTE: The number of cycles will depend on the cell type, RNA concentration, RNA integrity. The goal is to use the least number of cycles to obtain ~500pg/ μ L of cDNA. This will ensure as little amplification bias as possible while having enough cDNA for the subsequent steps. One strategy would be to perform SMART-Seq with 20-50% of your purified RNA with a high number of cycles (i.e. 16 cycles) and then calculate the number of cycles needed from there.

NOTE: If you collected the gene-negative and gene-positive populations and you would like to know if you collected the correct cell types, you can run qPCR or ddPCR with cell type specific markers at this stage.

59. Use 150 pg of cDNA for Nextera XT library kit.

NOTE: HS DNA BioAnalyzer chip should be run on the BioAnalyzer after the SMART-Seq v.4 protocol and after Nextera XT indexing to ensure proper library construction.

Reagents

NOTE: Especially for the FISH/Probe Synthesis Reagents, use only the reagents with the catalog numbers listed below. We have not tested a wide range of reagents made by other companies.

NOTE: These volumes are per sample except the DMEM/BSA, which is for 5 samples. If performing Probe-Seq for more than one sample, adjust volume accordingly.

HBSS/FBS:

HBSS	900 μ L	(Thermo Fisher Scientific, cat. #14025092)
FBS	100 μ L	(Thermo Fisher Scientific, cat. #10437028)
Total Volume	1000 μ L	

DMEM/BSA:

DMEM	3000 μ L	(Thermo Fisher Scientific, cat. #11995065)
BSA	12 mg	(MilliporeSigma cat. #A9418)
Total Volume	3000 μ L	

Papain Mix:

HBSS	315 μ L	
1M HEPES pH 7.0	35 μ L	(Thermo Fisher Scientific, cat. #15630080)
50mM L-Cysteine	20 μ L	(MilliporeSigma, cat. #168149) **Dissolve 15 mg of L-Cysteine in 2.5 mL of UltraPure water - make fresh
0.5M EDTA pH 8.0	0.4 μ L	(Thermo Fisher Scientific, cat. #AM9260G)
Nuclease Free Water	19.6 μ L	
Papain	10 μ L	(Worthington, cat. #LS0003126)
Total Volume	400 μ L	

4% PFA (Thermo Fisher Scientific, cat. # 28908)

NOTE: Make 4% PFA from the 16% PFA ampule by diluting in PBS - make fresh before experiment.

Permeabilization Buffer:

PBS	989 μ L	
Hoechst 33342 (10mg/mL)	1 μ L	(Thermo Fisher Scientific, cat. #H3570)
10% Triton X-100	10 μ L	
Total Volume	1000 μ L	

FACS Collection Solution (1% BSA/PBS):

BSA	10 mg
PBS	1000 μ L
Total Volume	1000 μ L

NOTE: Make the above solutions fresh before the experiment.

NOTE: Keep HBSS/FBS and DMEM/BSA at RT and 4% PFA and Permeabilization Buffer on ice.

NOTE: The papain mix should be incubated at 37°C for 10 minutes before use. It will be cloudy initially and become clear after warming.

NOTE: Add 1 μ L of RNasin Plus (Promega, cat. #N2615) for every 1 mL of every solution used. Incubate for at least 10 minutes with the RNasin before use.

FISH Reagents

The details of the SABER reagents can be found in Kishi, Lapan, Beliveau, West et al., (2019) *Nature Methods* paper. For detailed protocol for gene-specific probe set design, see the saber.fish website. Here are the reagents that you will need for Probe-Seq:

40% wHyb:

20x SSC	5 mL	(Thermo Fisher Scientific, cat. #15557044)
UltraPure Water	25 mL	
Deionized formamide	20 mL	(MilliporeSigma, cat. #S4117) *Aliquot and store at -20°C
Total Volume	50 mL	

Store at -20°C. This solution should not freeze at -20°C.

Pre-warm to 43°C before use.

NOTE: Compared to the original SABER-FISH paper (Kishi et al., 2019), these solutions do not contain Tween-20. This is because we have found that Tween-20 causes cell clumping.

Hyb1:

20x SSC	1 mL	
UltraPure Water	1 mL	
Deionized formamide	4 mL	
50% Dextran sulfate	2 mL	(MilliporeSigma cat. #D8906) *Dissolve powdered dextran sulfate in water overnight to make 50%. Stored at -20°C.
Total Volume	8 mL	

Store aliquots of 120 µL at -20°C. This solution should not freeze at -20°C

Probe Mix:

Hyb1	96 µL
Purified Probe set	1 µg
UltraPure Water	up to 120 µL

Pre-warm to 43°C before use.

Fluorescent Oligo Mix:

PBS	100 µL
10 µM Fluorescent oligo	2 µL (each)

Pre-warm to 37°C before use.

NOTE: Fluorescent oligos are ordered from IDT. The sequences and the fluorescent tags (on the 5' end) can be found in either the SABER (Kishi et al., 2019) or Probe-Seq (Amamoto et al., 2019) paper. They are resuspended in UltraPure water as 100 uM stocks, and 10 uM working stocks (diluted in water) can be aliquoted and stored at -20°C.

NOTE: Add 1 µL of RNasin Plus (Promega, cat. #N2615) for every 1 mL of every solution used. Incubate for at least 10 minutes with the RNasin before use.

Probe Synthesis

NOTE: Brian Beliveau has previously published Genome-wide probe sets for *C. elegans*, *D. melanogaster*, *H. sapiens*, *A.thaliana*, *M. musculus*, and *S. danios*. If you would like to use other species, please refer to OligoMiner to generate your genome-wide probe set.

Before you start:

- Get access to your institution's computing cluster.
- Create a folder on the cluster (i.e. "SABER_Probe_Design").
- Download Brian Beliveau's OligoMiner-master folder from Github: <https://github.com/brianbeliveau/OligoMiner>
- Transfer the OligoMiner-master folder in the SABER_Probe_Design folder on the cluster. I usually use rsync command in the Terminal (Mac) to transfer files between the cluster and the local computer (example shown below).
- Download the Genome-wide probe set for your favorite organism from Ting Wu's OligoPaint website: <https://oligopaints.hms.harvard.edu/genome-files>
- Download the "Complete Genome" with the "Balance" setting. Unzip the file and transfer the folder in the SABER_Probe_Design folder on the cluster.

Obtaining the gene-specific BED file (30 minutes)

1. Go to the UCSC Genome Browser: <https://genome.ucsc.edu/>
2. Click on Genomes and go to the genome browser of your organism (i.e. GRCm38/mm10 for mouse).
3. Enter the gene of your choice in the search box (i.e. Grik1), click on the correct mRNA transcript in the drop-down menu (i.e. Grik1 (Mus musculus glutamate receptor, ionotropic, kainate 1 (Grik1), transcript variant 1, mRNA. (from RefSeq NM_146072)), and hit Enter.
4. With the correct mRNA in the viewer, hover the cursor over Tools, and click on Table Browser in the drop-down menu.
5. Fill in as follows:
 - Group: Genes and Gene Predictions
 - Track: NCBI RefSeq
 - Table: UCSC RefSeq (refGene)
 - Region: position
 - Output format: BED
 - File type returned: plain text
6. Name the output file (i.e. Grik1_Mm).
7. Click on Get Output.
8. Click on "Exons plus 0 bases at each end".
9. Click "get BED". This will download your BED file.
10. Open the BED file. I use a software called Brackets, which is free to download at: <http://brackets.io/>
11. If there are multiple isoforms listed (i.e. multiple NM_# with different #), choose one of the isoforms to keep and manually delete the others. Save the file.
12. Note 2 features in the BED file as you will need them later:
 - a. The chromosome number
 - b. Strand orientation "+" or "-"
13. Transfer the BED file to the SABER_Probe_Design folder on the cluster (i.e. type "rsync Grik1_Mm user@o2.hms.harvard.edu:/home/user/SABER_Probe_Design" in the Terminal (Mac))
14. Log into the cluster.

NOTE: How you log into your cluster will be different institution to institution.

15. Run an interactive mode (i.e. “srun --pty -p interactive --mem 500M -t 0-06:00 /bin/bash”).
16. Load these modules:
 - a. conda2 (i.e. type “module load conda2”)
 - b. bedtools (i.e. type “module load bedtools/2.27.1”)
 - c. Any other dependencies (i.e. gcc)

NOTE: The versions will be different for every cluster.

17. Run intersectBed between the genome-wide probe set (using the correct chromosome BED file) and your BED file with the option of -f 1. (i.e. Type in “intersectBed -a mm10b/mm10_chr21b.bed -b Grik1_Mm -f 1 > Grik1_Mm+_probes.bed”)

NOTE: Make sure you're running this command in the folder with the genome-wide probe set (i.e. mm10b) and your gene-specific BED file (i.e. Grik1_Mm).

NOTE: If you have enough computing power, it's also possible to run this command locally.

18. If your gene-specific BED file had a “+” orientation, you need to reverse complement. Use the probeRC.py file from the OligoMiner-master folder (i.e. Type in “python OligoMiner-master/probeRC.py -f Grik1_Mm_probes.bed -o Grik1_Mm-_probes”)
19. Transfer the finalized gene-specific probe set file to the local computer (i.e. using the Terminal, be in a folder on the local computer, and type “rsync user@o2.hms.harvard.edu:/home/ra108/SABER_Probe_Design/Grik1_Mm-_probes.bed .”)
20. Open the gene-specific probe set file (i.e. using Brackets) and see how many oligos you have.

NOTE: Sufficient number of oligos is crucial for successful Probe-Seq. We have used as few as 12 oligos for high expressing transcripts and saw a nice separation by FACS. However, in general, more is better. For low expressing transcripts, if fewer than <20 oligos, Probe-Seq may not work well.

Oligo and hairpin ordering (10 minutes + wait time for synthesis and delivery)

21. For ordering the oligos, we use IDT. Oligos from other companies have not been tested.
22. Go to www.idtdna.com and under “Products & Services” click on “Custom DNA oligos.”
23. Under “DNA oligos,” click on “Order Now” and “All ordering options.”
24. Under “Ordering,” click on “Plates” and order 25 nmole DNA Plate oligo in a 96 well format.

NOTE: If you contact your IDT sales rep, you should be able to get an option of 10 nmole DNA plate oligo, which further decreases the cost of oligos.

25. Click on “Upload Plates” and download the sample ordering template.
26. Open the template with Excel.
27. Copy and paste the contents of the gene-specific probe set file into a region of the Excel template that is not being used.
28. Copy the oligo sequences into the column under “Sequences.” Delete the other pasted information.

29. For each oligo sequence, add “TTT” and a primer sequence. If using two genes for one sort, two different hairpin primer sequences should be added. From my experience, the hairpin primer sequences that have worked well are:

25.25 CCAATAATA
27.27 CATCATCAT
28.28 CAACTTAAC
36.36 AACTAATCT

NOTE: Other hairpins have been used by our group and others. However, these are the hairpins that have worked most consistently for me. For other hairpins, please see the original papers.

30. For each line, the sequence should read, “(oligo sequence)TTT(primer sequence)”

NOTE: You can find examples of sequences in Supplementary Files of the Probe-Seq paper.

31. Name each line with the name of the gene, hairpin used, and a number (i.e. Grik1_25_1)
32. Many genes can fit on one plate, but it's advisable that a new gene starts in a new row every time (i.e. If Grik1 oligos use 30 wells, and thus 3 rows – 2 full rows + 6 wells - the Grm6 oligos should start on the 4th row)
33. Upload the Excel file on IDT.
34. Under Plate Specification, use the following:
- Scale: 10 nmole (or 25 nmole)
 - Purification: Standard Desalting
 - Plate Type: V-Bottom
 - Ship Option: Wet
 - Buffer: IDTE 7.5 pH
 - Normalization Type: Normalized Yield
 - Quantity: 5 nmol
 - Concentration: 100 µM
 - Volume: 50 µL
35. Order plate.
36. When the plate arrives, thaw the DNA, and use a P200 multichannel pipette to pipette the oligos into a sterile reservoir to pool the oligos. The pooled oligos can be stored at -20°C.
37. Make a working stock of the pooled oligos (10 µM) by diluting the 100 µM stock with IDTE pH 7.5 (IDT Catalog #11-05-01-05).

NOTE: Do not use other TE buffers.

NOTE: The working stock can undergo freeze-thaw multiple times.

38. Order hairpins as written in the Supplementary Files of the SABER (Kishi et al., 2019) or Probe-Seq (Amamoto et al., 2019) paper. They can be ordered dry and resuspended in IDTE pH 7.5 as 100 µM stock solution. Working stock (5 µM) can be made using IDTE pH 7.5 and aliquoted for further use.

NOTE: The 25 hairpin requires a InvdT at the 3' end and HPLC purification.

Probe set synthesis (2-3 hours)

39. Add the following reagents in a PCR tube on ice:

UltraPure Water	44.5 µL
10x PBS	10 µL
100mM MgSO ₄ (NEB)	10 µL
dNTP(A,C,T 6mM each, NEB)	5 µL
Clean.g (1µM)	10 µL
BST enzyme (McLab)	0.5 µL
Hairpin (5µM)	10 µL

NOTE: 6 mM A, C, T dNTP is made by mixing dATP, dCTP, and dTTP to a final concentration of 6 mM.

NOTE: Clean.g is ordered through IDT with the following sequence (standard desalting, diluted with UltraPure water):

CCCCGAAAGTGGCCTCGGGCCTTTTGGCCCGAGGCCACTTTTCG

40. In a thermocycler, incubate for 15 minutes at 37°C.

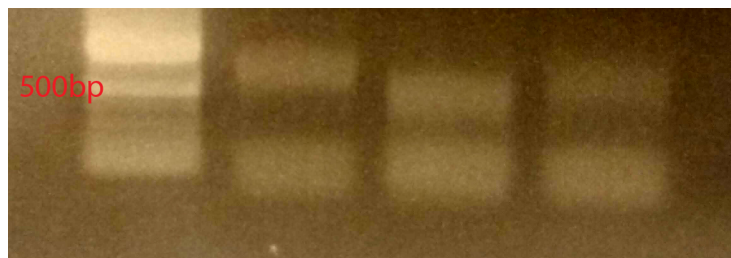
41. Add 10 µL of the gene-specific oligo pool (10 µM) and mix.

42. Incubate for 100 minutes at 37°C, incubate for 20 minutes at 80°C, incubate indefinitely at 4°C. You can store this unpurified reaction at -20°C.

43. Check the length of the extended probe set by taking 8 µL of the unpurified probe set and mix it with 1.6 µL of 6x Loading Dye (or any other Loading Dye should be fine). Run a 1.25% ethidium bromide agarose gel at 150V for 8 minutes. The band should be between 300 – 700bp. See below for an example.

NOTE: Longer running time will lead to fainter band. Two bands are expected – the top band is your extended probe and the bottom band contains catalytic hairpins and the clean.G oligo. Even after purification, this bottom band remains and may even help stabilize the long ssDNA probes during storage.

NOTE: If the probe set is too short or too long, you can change the extension time from 100 minutes to 80 minutes (if too long) or 120 minutes (if too short). The hairpin amount can also be changed. The hairpin amount can also be changed. For example, if you used 10µL of hairpin and your probe runs at ~250bp, you can try doubling the hairpin to 20µL (and decrease the amount of water in the reaction) to double the length to 500bp.



44. Purify the remaining 92 µL of probe set using a MinElute PCR Purification Kit (Qiagen cat. #28004). We use 7x PB buffer (instead of 5x) to increase binding efficiency.

45. Elute in 25 µL of UltraPure water.

46. Determine the concentration by NanoDrop using the ssDNA setting. The concentration should be ~200 – 400 ng/ µL.

47. Store purified probe set at -20°C.

Troubleshooting Guide

Issue	Possible Solutions
The probe doesn't extend	• Make sure that all of the solutions have been diluted in the correct solutions (water vs. IDTE pH 7.5 vs. PBS). • Check the ordered oligo sequences that the "TTT" linker and the hairpin primer sequences have been added. • During probe synthesis, make sure that the hairpin primer on the oligos and the hairpins match (i.e. Grik1.25 and 25.25 hairpin).
There's no separation between gene-positive and gene-negative events	• Make sure that the marker gene is expressed highly. • Check the number of oligos. If less than 15-20, there might not be enough tiling oligos, and a new marker gene may be necessary. • Make sure that the temperature of the solution does not deviate from the appropriate temperature (i.e. 43°C) during solution changes.
Cells are clumped together	• Make sure you do not use Tween-20 in your wHyb or Probe Mix. • Dissociate the tissue for longer. • There may be other dissociation methods for your tissue of interest.
The signal is not strong enough to separate the populations	• If the number of oligos is low, you can try using both Exons and Introns when generating your probe set. This has not been tested in our hands, but it should increase signal. • Use more probe (2 µg compared to just 1 µg). • Incubate for longer time (24 hours instead of 16 hours).

Appendix: Nuclei Isolation

NOTE: All of the FISH reagents remain the same as the whole cell protocol. Before beginning, make sure to add RNase inhibitor to every solution. Also, before beginning the nuclei isolation, make sure to pre-warm the reagents. 40% wHyb (warm 500 μ L per sample) and Probe Mix should be warmed to 43°C.

NOTE: Because you are using only nuclear RNA as the marker, the signal intensity is likely to decrease compared to whole cell. Therefore, the expression level and transcript length of the marker gene will be more critical.

NOTE: If not enough signal is detected, you can design the oligos with both Exon and Intron sequences, which should be present in nascent nuclear RNA.

NOTE: The total extracted RNA is less abundant compared to whole cell. Therefore, when performing SMART-Seq v.4, more cycles will be needed to amplify your cDNA.

Nuclei Isolation (30 minutes)

NOTE: This nuclei isolation method works for mouse and human retina and brain. It has not been tested for other tissues in other organisms.

NOTE: This nuclei isolation is meant for frozen tissue. If you have fresh tissue, flash-freeze the tissue in isopentane/dry ice slurry. Keep the collection tubes on dry ice before starting. The tissue goes directly into the slurry, without too much residual liquid around the tissue. Quickly transfer the frozen tissue into the collection tube. Keep at -80°C.

1. Prepare all solutions and keep on ice with the Dounce homogenizer (i.e. Thomas Scientific cat. #1234F37).
2. Fill the glass Dounce homogenizer with 1 mL of cold homogenization buffer. Keep on ice.
3. Mince the tissue into little pieces and place in 1% PFA for 5 minutes on ice.
4. Transfer the tissue pieces into the homogenizer with the buffer. With the tight pestle, homogenize with 10-15 strokes on ice. Avoid foaming.
5. Transfer the homogenate into a 5-mL polypropylene tube (Thermo Fisher Scientific cat. #14-959-11A). Make sure to seal the cap by pushing in.
6. Carefully add 2 mL of Sucrose Bed solution to the bottom of the tube so that the homogenate is above the sucrose solution.
7. Spin at 500xg, 12 minutes, 4°C.
8. Remove supernatant. Add 1 mL of 4% PFA solution and resuspend pellet. Incubate for 15 minutes at 4°C with rocking.

NOTE: Remove the top layer supernatant first. Then, remove the rest of the supernatant.

9. Centrifuge at 2000xg for 5 minutes at 4°C.

-----GO TO STEP 17 of the Probe-Seq Protocol-----

Nuclei Isolation Reagents

BUFFER 1:

1.5M Sucrose	2500 µL
1M KCl	375 µL
1M MgCl ₂	75 µL
1M Tris Buffer pH 8.0	150 µL
Nuclease-Free Water	11900 µL
Total Volume	15000 µL

BUFFER 1 can be stored at 4°C for up to 6 months

Homogenization Buffer:

BUFFER 1	968 µL	
10% Triton X-100	10 µL	
Protease Inhibitor (50x)	20 µL	(Promega, cat. #G6521) *Reconstituted in DMSO
1 mM DTT	1 µL	
Hoechst 33342	1 µL	(Thermo Fisher Scientific, cat. #H3570)
Total Volume	1000 µL	

Homogenization Buffer should be made fresh

NOTE: 1 mL of Homogenization Buffer is necessary for each sample.

SUCROSE BUFFER:

1M KCl	2250 µL
1M MgCl ₂	450 µL
1M Tris Buffer pH 8.0	900 µL
Nuclease Free Water	11400 µL
Total Volume	15000 µL

SUCROSE BUFFER can be stored at 4°C for up to 6 months

Sucrose Bed:

SUCROSE BUFFER	2500 µL
24% Sucrose	12500 µL
Total Volume	15000 µL

Sucrose Bed can be stored at 4°C for up to 6 months

****Add 1 µL of RNasin Plus (Promega N2615) for every 1 mL of every solution used at least 10 minutes before use.**